

service provider, since the network is already in place and more widespread than ever. In developing countries, the electric power line infrastructure is more widespread and robust as compared to the telephone network, making it possible to connect even the remotest location to the Internet.

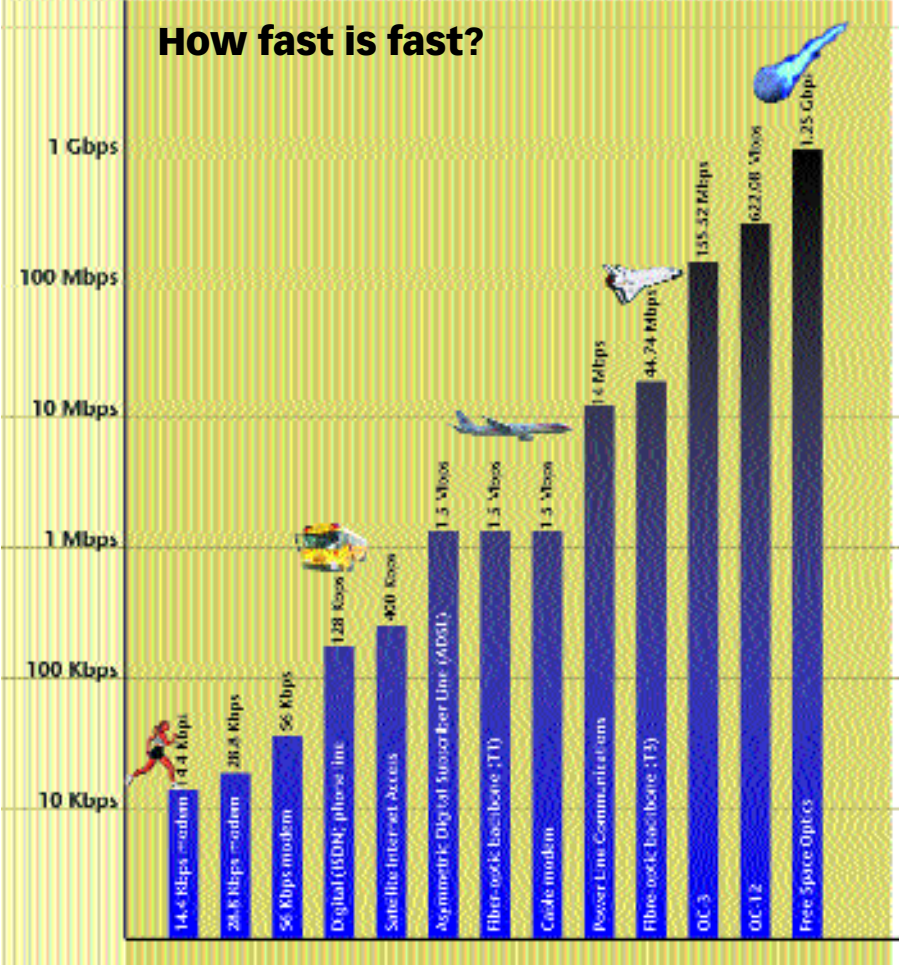
Thomas Edison once said, "I'll make electric light so cheap that only the rich can afford candles." PLC is all set to deliver on a similar promise for Internet connectivity.

The Web from outer space

Satellite communication has been here for a while; cable television transformed our television watching experience from a bland three-channel serving to a splendid buffet of over 150 channels! These silent machines set in the blackness of space are now entering the realm of home Internet access.

Satellite Internet Access (SIA) is a relatively high-speed service that uses orbiting geo-synchronous satellites to beam down information. However, this technology works in tandem with your existing Internet connectivity; users still connect to their present Internet service provider to upload and transmit data.

The process is very simple—you dial into your Internet service provider and the requested data is sent from your ISP to a special server from the SIA provider called the Network Operations Centre (NOC). This server retrieves the requested information from the Web server but instead of sending it back through the ISP, it beams it up to an orbiting satellite, which in turn transmits the data to a little satellite dish out-



There's been a quantum jump in last-mile Internet connectivity. This logarithmic graph depicts the bandwidth of various Internet technologies from the lowly modem to FSO.

side the user's location. From here, it passes through a special modem that decodes it and sends the information to the user's computer. The entire process takes less than half a second to complete, which is rather long for a single request of information.

Due to this inherent latency, SIA is most suited to applications involving sustained data transfer (such as large file downloads and streaming audio and video applications) rather than repeated access to small amounts of data (as in Web browsing or playing games). Another disadvantage is that it is still expensive and relatively more difficult to install than DSL or cable. The service promises download speeds up to 400 Kbps, while the upload speeds depend upon the service provider.

From Fiction to Fact

Arthur C. Clarke, the science-fiction novelist, first postulated the idea of using satellites for point-to-point radio communication back in 1947

However, newer systems offering download speeds up to 1.5 Mbps, with upload speeds of 200 Kbps are in the pipeline. Understandably, this system, in its present state, will be more suited to applications that stress on downloading information rather than uploading.

What tomorrow holds

Yesterday was the era of text Internet; today is the age of sound and animation; tomorrow will usher in a radical epoch of high-quality audio and video streaming, interactive video-conferencing and megabyte downloads in minutes. The last mile has always been a hurdle in the path of high-bandwidth Internet connectivity. Once these new technologies are economically viable, our perception of the Internet will change forever. And the dice has already been rolled.

MARCO D'SOUZA

Web Resources

Free Space optics
<http://www.terabeam.com>
<http://www.lightpointe.com>
<http://www.lightpointe.com/index.cfm?fuseaction=technology.WhitePapers>
http://www.airfiber.com/tech/WP_laser_eye_safety.pdf
<http://www.freespaceoptic.com/>

Power Line Communications
<http://www.homeplug.org>

Satellite Internet Access
http://www.isp-planet.com/fixed_wireless/research/satellite.html